Empirical observations

4

Cifar-10 dataset:

ResNet-18 (with Batch Norm):

	Train accuracy	Test accuracy
GD	100 %	~72% (??)
SGD	100 %	85 %
SGD + momentum	100 %	89 %
SGD + mom + augm + ℓ_2	100 %	95 %
GD + mom + augm + ℓ_2	100 %	87 %

Stochastic Training is Not Necessary for Generalization, Geiping et al. 2021
Bad Global Minima Exist and SGD Can Reach Them, Liu et al. 2020
Revisiting Small Batch Training For Deep Neural Networks, Masters and Luschi 2018

Neural networks perform well in practice

(even without regularisaton)

0 training loss

Good generalisation

(overfitting?)

Empirical observations

Cifar-10 dataset:

ResNet-18 (with Batch Norm):

	Train accuracy	Test accuracy
GD	100 %	~72% (??)
SGD	100 %	85 %
SGD + momentum	100 %	89 %
SGD + mom + augm + ℓ_2	100 %	95 %
GD + mom + augm + ℓ_2	100 %	87 %

Neural networks perform well in practice
(even without regularisaton)

Stochastic Training is Not Necessary for Generalization, Geiping et al. 2021
Bad Global Minima Exist and SGD Can Reach Them, Liu et al. 2020
Revisiting Small Batch Training For Deep Neural Networks, Masters and Luschi 2018

0 training loss
Good generalisation
(overfitting?)

Main questions

Mystery of training neural networks with (stochastic) gradient descent

- Do they converge to 0 training loss? — non-convex optimisation
- Why do they generalise? — overfitting $\neq$ bad generalisation?
- What is the role of initialisation? — important since non-convex

In this talk:

- 2 layers neural networks
- (Leaky) ReLU activation
- Gradient Flow
- No regularisation